Natural Experiments in the Public Record of AI: A Systematic Survey

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July 2026
analysis passes of record: 2026-07, `natex` v0.2.0; all stochastic steps seeded

Abstract

The public record of AI — benchmark scores, adoption surveys, capital expenditure, chip stocks, model registries, crowd-sourced battle streams — is full of dated events that are claimed, usually by eye, to have bent some curve. We survey a systematic attempt to test such claims with formal quasi-experimental designs, run and validated by the open-source `natex` toolkit: 17 dataset–design analyses carried to a verdict and 15 more triaged and excluded for cause, each subjected to a mandatory falsification battery (specification grids, shifted-cutoff placebo grids, placebo units and outcomes, density and covariate checks, pinned controls, honest refusals). Eight findings are selected here by credibility and interest. Three are credible: the 2025 BTOS questionnaire rewording moved measured US business AI adoption by +6.0 pp (4.7× the largest of 49 placebos), the October-2023 export controls bent China's legal AI-chip stock by −0.3 to −0.56 log-units/yr against every control group (25/25 specifications, loophole-round placebo null), and GPQA-Diamond's slope 2.4×'d at o1-preview with a clean placebo grid (0/7). One is moderate: an 83% post-Act deficit of models just above the EU AI Act's 10^{25}-FLOP line. Two are first-class negatives: the Epoch Capabilities Index shows no kink at o1 in a design with demonstrated power, and the sycophantic GPT-4o build won zero opponent-adjusted votes on LMArena. And two are cautionary: a sharp January-2025 adoption acceleration that cannot be attributed to DeepSeek-R1, and a hyperscaler capex "kink" at ChatGPT whose nominal $p = 4 \times 10^{-6}$ collapses to a placebo-calibrated 0.125. Across the corpus, roughly half of the nominally significant estimates die in the battery — on short, serially dependent, composition-churned series, the falsification battery is not a robustness appendix; it is the analysis.

1 Introduction

Public discussion of AI progress and AI policy runs on dated trend breaks. Reasoning models "changed the slope" of capabilities; ChatGPT "ignited" the datacenter buildout; export controls "choked" China's compute; DeepSeek "bent" the adoption curve; the EU AI Act "chilled" large training runs. Each claim names an event, a series, and a shape — which makes each one a candidate for a formal quasi-experimental design: a regression discontinuity or kink at a known cutoff [20, 35, 13], a difference-in-differences or synthetic control around a dated policy [2], bunching at a statutory threshold [71, 59, 58], or an interrupted time series with reversal [9]. Treating the claims this way buys an explicit identifying assumption, a standard error, and — most importantly — a falsification battery that can *reject* them.

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This paper is the capstone of a project that did this systematically. The instrument is `natex` [50], an open-source toolkit for automated natural-experiment discovery and estimation in the lineage of the LoRD3 local-discontinuity scan of [38, 37, 55], extended with kink and difference-in-kinks estimators following [13], a subset-scanning panel DiD, synthetic controls, and density/bunching tests. Over successive analysis passes the project pointed this machinery at the public quantitative record of AI — Epoch AI's model, benchmark, and chip databases [26, 28, 27], the Census Bureau's Business Trends and Outlook Survey (BTOS) [14], SEC EDGAR filings, LMArena's published battle stream [23], and semiconductor market data — carrying 17 dataset–design pairs to a recorded verdict and triaging 15 more as infeasible or ill-posed (Section 11 tabulates all 32). Each completed analysis is written up as a companion note with committed figure scripts that re-derive every headline number from the frozen inputs before drawing [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49].

Here we select the eight strongest results by the product of credibility and interest, and — deliberately — count honest nulls and identified artifacts as first-class findings alongside the positive estimates. The selection spans the full verdict taxonomy the project converged on: *credible* (the estimate survives the battery), *descriptive-only* (the data feature is real but attribution to the named event fails), *null-with-power* (the dial reads zero where the test demonstrably can move), and *identified artifact* (the battery caught the machinery, or the platform, generating the result). Section 2 covers methods, Sections 3–10 the findings, Section 11 coverage, and Section 12 the lessons.

2 Methods: the toolkit and the validation philosophy

Design families. `natex` v0.2.0 implements, behind one CLI: local-polynomial regression discontinuity with the LoRD3 blind discontinuity scan and honest (split-sample) post-discovery 2SLS [38]; sharp and fuzzy regression kink and difference-in-kinks (DiK) estimators [20, 13]; a subset-scanning panel difference-in-differences ("SuDDDS") that searches units and break dates by likelihood-ratio scan, calibrated by permutation under dependence-preserving nulls; synthetic-control and related panel estimators [2, 1]; McCrary-style binned-Poisson density tests [63] used both as manipulation diagnostics and as bunching estimators; and event-study/interrupted-time-series segmentations [9]. House rules bind every run: a single seeded random generator threaded through all stochastic calls, discovery steps that never read the outcome, and failed computations that return NaN — a refusal — rather than a number.

The falsification battery. No estimate in the corpus is reported without its battery. The standing components: (i) specification grids over bandwidth, kernel, donut, and polynomial degree; (ii) *shifted-cutoff placebo grids* in the spirit of [32], read as separating bend *existence* from date *attribution* — rejections at non-event dates mean the series bends over an era or everywhere, not at the event; (iii) placebo-in-space (pseudo-treated units) and placebo outcomes; (iv) density and covariate discontinuity checks; (v) sibling-series falsifications — an aggregate in which the test demonstrably has power but should read null; and (vi) where available, *physically pinned controls* — units that cannot have been treated, whose movement identifies platform-side confounds (Section 8). Nominal HC1/CR1/HAC inference is reported but never trusted on its own: on short, serially correlated aggregates conventional robust standard errors are known to be badly oversized [10, 19, 35], so placebo-calibrated p-values and permutation ranks are the inference of record wherever the battery can supply them.

Table 1: The eight selected findings.

Finding (section)	Design / cutoff	Headline	Verdict
Export controls vs. China's compute (§3)	group DiK, chip stocks, 2023-10-17; SuDDDS DiD, big runs, 2022Q4	legal stock -0.56 ln/yr ($t = -3.3$; 25/25 specs); total stock $\approx$ half, fragile; big-run DiD +3.2/qtr fails placebo-in-space	credible / attenuated / descriptive
BTOS question rewording (§4)	measurement RDiT, wave 202524 (2025-11-03)	+6.04 pp, HAC $z = 16.3$; 4.7× max of 49 placebos; blind scan re-localizes	credible
EU AI Act 10^{25}-FLOP line (§5)	bunching / density, statutory threshold, post 2024-08-01	Fisher OR 0.173, $p = 0.007$; 82.7% missing mass; $\theta < 0$ in 15/15	credible (moderate)
GPQA-Diamond at o1-preview (§6)	sharp RKD-in-time, 2024-09-12	+0.00258 logit/day ($t = 3.3$); placebo size 0/7	credible, date-localized
US adoption at DeepSeek-R1 (§7)	group DiK + SuDDDS, BTOS sectors, 2025-01-20	DiK +7.3 pp/yr ($\lvert t\rvert = 2.6$); scan localizes exactly; placebo dates as large	descriptive-only
GPT-4o sycophancy episode (§8)	ABA ITS + Bradley–Terry, deploy 04-25, rollback 04-29	deploy style break real (perm $p = 0.013$); BT vote gain +0.02 log-odds (se 0.14): null	credible null / identified artifact
Hyperscaler capex at ChatGPT (§9)	group DiK, EDGAR panel, 2022-11-30	+0.140 dex/yr, nominal $p = 4\times10^{-6}$; placebo-calibrated $p = 0.125$	identified inference artifact
ECI at o1, fresh vintage (§10)	sharp RKD-in-time, 2024-09-12	+0.0065 pts/day ($t = 0.50$); placebos reject 4/8 elsewhere	null with power

Validation anchor. Before being trusted on novel data, the pipeline was run blind on the canonical Abadie–Diamond–Hainmueller Proposition-99 panel [2]: all three SuDDDS scan methods recover (California, 1989) exactly (precision = recall = 1.0), the deterministic simplex weights put summed weight 0.955 on ADH's five published donors, and the synthetic-control ATT of -19.5 packs per capita matches the canonical ≈ -19 average gap; the in-space RMSPE-ratio placebo ranks California 3/39 ($p = 0.077$) without ADH's poor-pre-fit exclusion, reported as the null it is [48]. The exercise has zero novelty by construction; its value is credibility transfer.

Audit lineage. Every analysis pass was produced under a two-stage protocol: an analysis agent runs the designs and freezes a numbers record; an independent audit pass re-derives the headline estimates from the frozen inputs, and the committed figure script of each companion note asserts every quoted number against that record before drawing. Audits are deliberately model-diverse after an early recorded failure in which same-family verifiers shared a PDF-extraction blind spot (dropped superscripts). Refusals, degenerate auto-configurations, and underpowered nulls of the automated pipeline are reported verbatim in each note ("stated as run"), not patched over.

3 Export controls and China's compute: three legs of one story

The October-2023 US export controls are the corpus's strongest policy result because the design decomposes the question [46]. *Leg 2* — the legal channel — is a group difference-in-kinks: the export-license schedule bends for China only at a common dated cutoff, so treated = China's legal

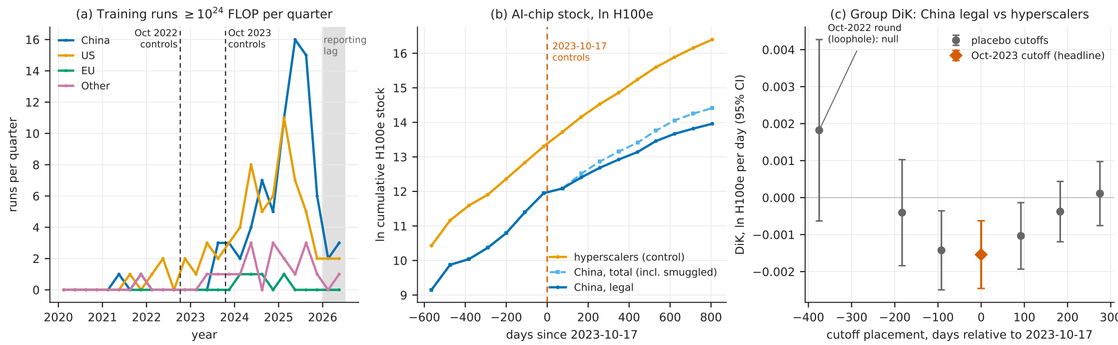

Figure 1: The three legs (reproduced from [46]): (a) quarterly $\geq 10^{24}$-FLOP training runs by country group; (b) ln cumulative H100e chip stock — hyperscaler controls, China legal, China total including smuggled — around the 2023-10-17 cutoff; (c) group-DiK estimates at the true cutoff and six placebo placements, with the Oct-2022 loophole round null at -375 days.

H100-equivalent chip stock, control = the four US hyperscalers, and the controls difference out the global supply bend. The headline DiK is -0.00154 ln-units/day (HC1 se 0.00047, $t = -3.30$; ≈ -0.56 ln-units/yr), negative in **25 of 25** specifications with a defensible band of -0.3 to -0.56 ln-units/yr. The falsifications land where the policy history says they should: the October-2022 first round — whose A800/H800 compliance loophole predicts no bite — is a null placebo ($+0.0018$, $p = 0.146$), and the placebo-treated group (Other vs. hyperscalers) is clean at the defensible bandwidths. *Leg 3* repeats the design on China's *total* stock including smuggled chips [56]: -0.00076 (se 0.00050, $t = -1.54$, 95% CI spanning zero) — half the size and fragile, because smuggled Nvidia compute grew from 27.8k to 662k H100e over 2024–2025 (doubling every 4.7 months) and now exceeds China's legal Nvidia stock 1.65×. *Leg 1* — a 4-country × 26-quarter DiD on counts of $\geq 10^{24}$-FLOP training runs — finds no suppression at all: the scan recovers {China} at 2022Q4 exactly (panel-null $p = 0.010$, the permutation floor), but the effect is *positive* (+3.2 runs/qtr, 95% CI $[+0.9, +5.5]$) and fails placebo-in-space at exact $p = 1.0$ — every pseudo-treated unit shows a larger studentized effect — so it is descriptive, not causal. The three-leg verdict: the controls bent the channel they legally govern, were substantially bypassed in aggregate, and left China's large-run training activity unbowed (Figure 1).

4 The question is the treatment: the BTOS rewording RDD

The cleanest single estimate in the corpus is a *measurement* natural experiment [41]. With wave 202524 (reference period from 2025-11-03; announced 2025-12-03), the Census Bureau reworded BTOS's headline AI question from use "in producing goods or services" to use "in any of its business functions" [21]. Treating the refresh as a known cutoff in a regression-discontinuity-in-time design on the spliced 71-wave national series [35], the jump is +6.039 pp (HC1 $z = 22.9$; HAC $z = 16.3$) against an old-wording level of 10.9 pp — a ratio of 1.553. The battery is uniformly supportive: all bandwidth, weighting, curvature, and HAC variants sit in a 6.0–7.5 pp band with $z > 12$; the largest of 49 placebo-cutoff estimates is 1.30 pp, 4.7× smaller (randomization $p = 0.020$, the 1/50 floor); and, told nothing, the LoRD3 scan's two top candidate boundaries exactly flank the known cutoff, with honest 2SLS +6.74 (95% CI $[6.01, 7.48]$). True adoption cannot produce it: the fitted pre-trend predicts +0.66 pp across the 50-day shutdown gap, and doubling that slope still leaves ≥ 5.38 pp. Nulls are reported as nulls (blind-scan randomization $p = 0.14$, underpowered at $n = 71$;

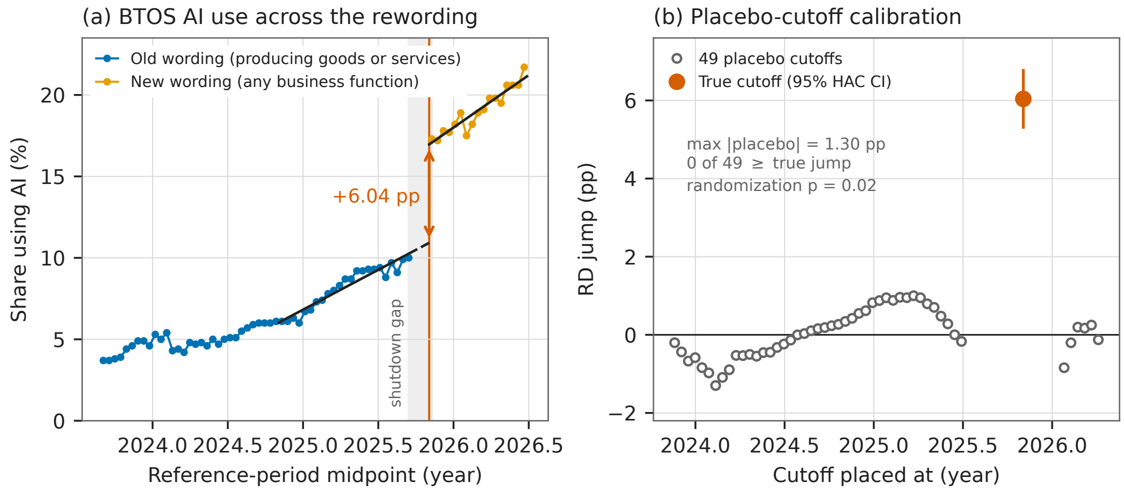

Figure 2: The rewording RDD (reproduced from [41]): (a) the spliced BTOS national series with per-side fits and the 50-day shutdown-gap extrapolation; (b) the same estimator at all 49 placebo cutoffs versus the true cutoff.

McCrary NaN — inapplicable on a uniform time grid). The estimate calibrates the splice factor — additive +6.04 pp, ratio 1.553 — needed to carry any old-wording BTOS analysis past December 2025, and implies that roughly a third of the post-refresh measured adoption level is attributable to asking a better question [12], with the caveat that "wording" bundles a simultaneous sample rotation and shutdown-adjacent composition change (Figure 2).

5 Bunching below the EU AI Act's 10^{25}-FLOP line

The EU AI Act's systemic-risk presumption at 10^{25} training FLOP [29] is a notch, and notches invite bunching [59]. In Epoch AI's compute panel (719 dated models with compute estimates), the post-Act density of new models just *above* the statutory line collapses [45]: a binned-Poisson density test at $\log_{10}$ FLOP $= 25.0$ is negative in 15/15 window × bin specifications (10/15 with $p < 0.05$), while the pre-Act period rejects in 0/15. The pre/post contrast is sharp: Fisher exact on below/above counts in the ±0.5-dex window gives OR $= 0.173$ (95% CI $[0.048, 0.619]$, $p = 0.0072$) — 28.9 post-Act models expected in $(25.0, 25.5]$ at the pre-Act ratio, 5 observed, an 82.7% deficit. The deficit is specific to the statutory line (placebo thresholds at 24.5 and 25.5 are null or opposite-signed, with mass piling up just *below* 25.0), specific to the post-Act period (a placebo split date inside the pre-Act sample gives OR 0.90, $p = 1.0$), and correctly timed (the above-line count collapses from 2025H1, after entry into force, before applicability), with the classic signature that marginal crossers vanish while the frontier jumps far above the notch (Grok 3, GPT-4.5, Grok 4 at $\log_{10} = 26.5$–26.7). Two facts grade it moderate rather than conclusive: only 5–13 post-Act models sit above the line (the preferred narrow-window interaction is a null, $p = 0.73$), and an Act-induced *disclosure* response — labs going quiet about compute near the line — would mimic bunching exactly. Either reading matters for the compute-threshold debate [73, 53]: the data just above the line already look threshold-aware, contrary to the threshold-blind scaling assumed in forward projections [60] (Figure 3).

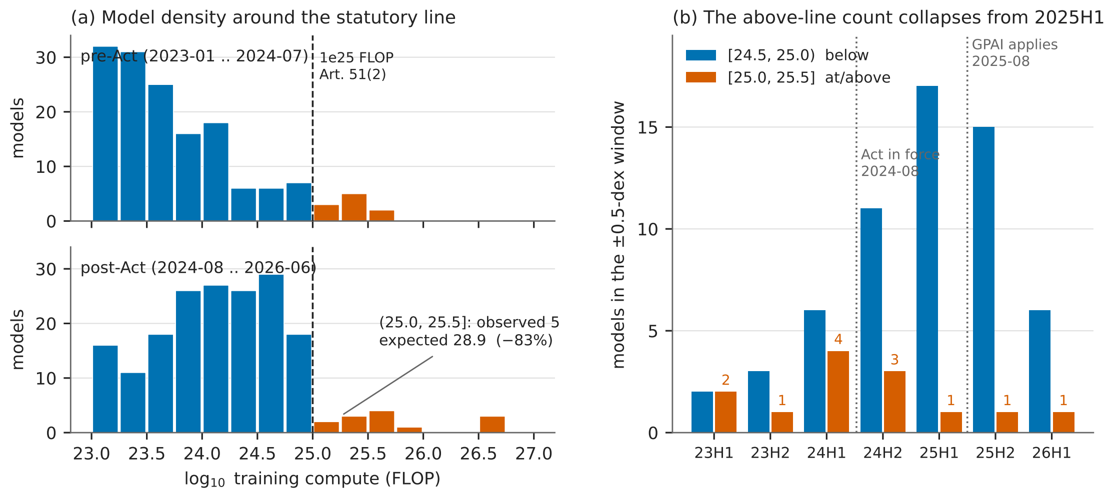

Figure 3: Bunching at the statutory line (reproduced from [45]): (a) model counts in 0.25-dex compute bins, pre- versus post-Act, around 10^{25} FLOP; (b) below/above counts per half-year — the above-line count collapses from 2025H1.

6 A date-localized capability kink: GPQA-Diamond at o1-preview

The flagship kink pass [39] supplies the survey's one date-localized capability result and its sharpest methodological contrast. GPQA-Diamond [70] logit-score slopes rise from 0.00188 to 0.00446 per day at the o1-preview release (2024-09-12) — roughly 15 → 34 raw points per year, a 2.4× acceleration. The headline RKD-in-time estimate is +0.00258 per day (se 0.00078, $t = 3.32$), positive in 8/8 bandwidth×donut cells, with an *empty* placebo grid (0/7 shifted cutoffs reject), a null release-density kink ($t = 0.76$), and a null training-compute covariate kink ($p = 0.54$): the bend localizes to the date (Figure 4). The contrast: METR's 50% time-horizon series [61] shows an equally real bend (doubling time 9.9 → 3.5 months, $t = 2.13$, positive in 8/8 cells) whose pre-side placebos *also* reject ($p = 0.004$–0.030 at −90 to −270 days) — a credible reasoning-era slope change that cannot be pinned to 2024-09-12. The pair is the era-bend/event-bend taxonomy in a single dataset family, and the standing caveat travels with both: composition — reasoning models entering the release stream — *is* the mechanism, so these are properties of the release stream, not statements that individual models improved.

7 A sharp bend, no verdict: US adoption at DeepSeek-R1

The project's flagship adoption analysis ends in an instructive split verdict [40]. In the BTOS sector panel (20 sectors × 54 biweekly waves), a sector-by-wave DiD scan localizes an acceleration in AI-exposed sectors (information; finance; professional services) at $t_0 = 2025.089$ — *exactly* the first survey wave after DeepSeek-R1's release on 2025-01-20 [33] — with scan $p = 0.010$–0.030; the exposed-minus-unexposed gap slope more than triples (+4.2 → +13.9 pp/yr), and the group DiK at the release date is nominally significant (+7.30 pp/yr, CR1 se 2.80, $|t| = 2.61$). The localization survives composition checks and placebo-in-space (0/13 pseudo-treated control sectors). But attribution to R1 fails five ways: placebo cutoffs at non-event dates return DiKs as large or larger (+11.4 pp/yr at $t = 2024.30$, $|t| = 3.21$; +4.0 at 2024.50, $|t| = 4.53$); only 2/6 specification variants at R1 are significant; equally large local kinks appear in non-exposed sectors (real estate,

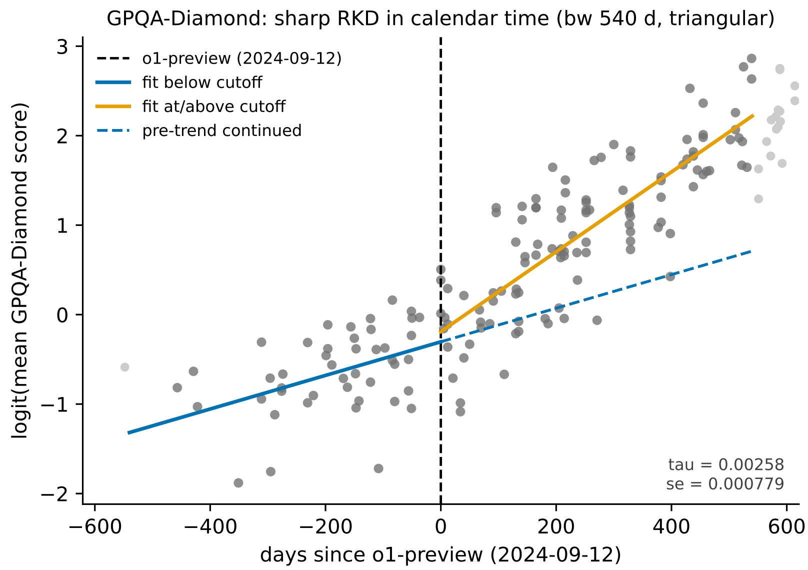

Figure 4: GPQA-Diamond, logit(mean score) against days since o1-preview (reproduced from [39]): per-side local-linear fits (bandwidth 540 days) versus the continued pre-trend; placebo empirical size 0/7.

arts, health care); the predicted *deceleration* at o1 sign-flips with bandwidth; and the cutoff week is confounded — the AI-diffusion export rule (2025-01-13), the Stargate announcement (2025-01-21), and a BTOS sample-year rollover in mid-December 2024 all sit inside the window. The verdict of record: something bent US business AI adoption in January 2025, and the localization is data-driven — but these data cannot say it was R1 (Figure 5).

8 Sycophancy did not win votes — and the rollback isn't what it looks like

The GPT-4o sycophancy episode of April 2025 [67] supplies the corpus's only deploy-and-rollback interrupted time series, inside 135,634 LMArena battles [47]. The *deploy* leg is credible: the live 4o alias shows a sharp markdown-style collapse timed to April 25–26 (headers per 1k tokens $3.52 \to 1.12$; sliding-cut permutation $p = 0.0128$, rank 1/78) while nine pinned dated snapshots stay flat — proving the arena copy was not pinned — and it delivers a precise null on the headline mechanism: the Bradley–Terry-adjusted win-rate change during the sycophancy window is $+0.02$ log-odds (SE 0.14; implied win rate $0.540 \to 0.545$). During the four days users saw the sycophantic build, it gained zero opponent-adjusted votes — a clean behavioral null on the preference-hacking mechanism the literature predicted [75]. The *rollback* leg is an identified artifact: on exactly 2025-04-29 the pinned `claude-3-7-sonnet-20250219` snapshot — whose weights cannot change — takes an equal-and-opposite persistent step in style *and* BT strength (-0.54 log-odds), demonstrating an arena-side serving change that contaminates every 29-April estimate. The 4o post-rollback regime never reverts (the episode is A–B–C, not A–B–A), and the widely repeated "sycophancy raised the win rate, rollback lowered it" narrative loads entirely on the confounded boundary. Pinned snapshots are the cheap, decisive placebo for platform time series (Figure 6).

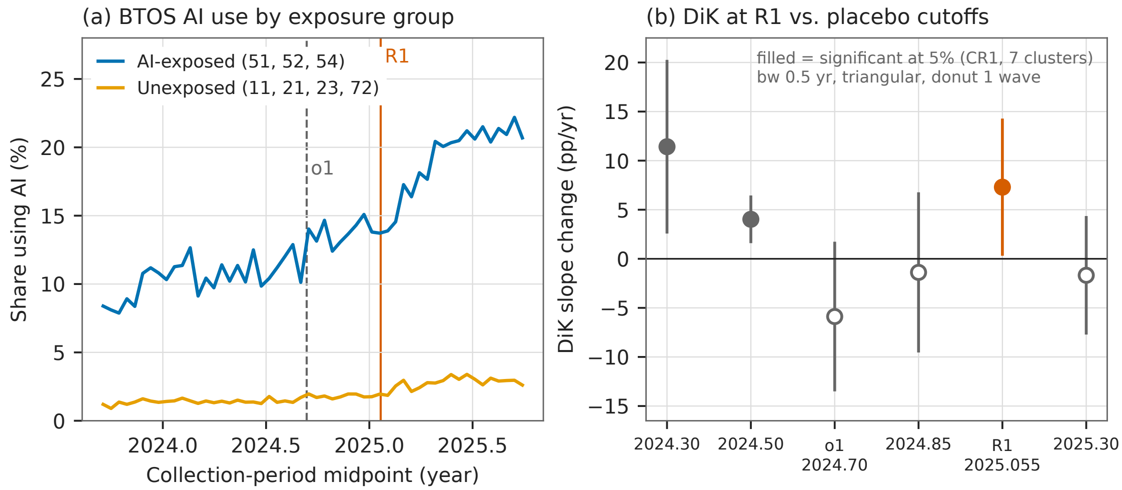

Figure 5: Adoption at R1 (reproduced from [40]): (a) BTOS AI-use share by exposure group with o1 and R1 marked; (b) group-DiK estimates at R1, o1, and four non-event placebo dates — two placebos are significant, one larger than the R1 estimate.

9 An identified inference artifact: hyperscaler capex at ChatGPT

The capex analysis is included as a first-class finding precisely because its battery caught the failure mode that would otherwise headline this literature [42]. Big-4 hyperscaler capital expenditure (EDGAR XBRL reconstruction) accelerates against a six-firm non-AI capital-intensive aggregate after ChatGPT: group DiK +0.1396 dex/yr (HC1 se 0.0302, nominal $p = 3.9 \times 10^{-6}$), positive in 18/18 specification cells, localized over 2023.0–2023.5 exactly as a lagged investment response would be, and visible per firm for Microsoft, Alphabet, and Amazon but null for Meta. Every conventional robustness check passes. What kills the causal reading is the clean pre-period placebo grid: at 15 non-event cutoffs in 2017–2020 the identical specification rejects at nominal 5% seven times (empirical size 0.47), the largest placebo $|z|$ (5.11, at 2019.75) exceeds the ChatGPT estimate's matched-bandwidth $|z|$ (4.35), a full-sample kink at the non-event date 2021.5 reaches $|z| = 6.9$, and the BIS export-control date eight weeks earlier is statistically indistinguishable (+0.125, $z = 4.0$). Placebo-calibrated, the ChatGPT estimate's p is 0.125–0.25: the nominal p-value overstates the evidence by five orders of magnitude, because HC1 inference assumes serial independence and the residual lag-1 autocorrelation reaches 0.66 [10]. The slope divergence is a genuine, robust data feature; the "significance" was an artifact of the standard errors, and only the battery could tell the two apart (Figure 7).

10 An honest null with power: the ECI at o1, fresh vintage

If "everything kinked in 2024," the per-benchmark results of Section 6 would be uninformative; the Epoch Capabilities Index (ECI) [51] is the aggregate guard, and it holds on a fresh vintage with twelve more months of data [44, 39]. The all-models RKD-in-time at o1-preview is +0.0065 ECI points/day (se 0.0131, $t = 0.50$, $p = 0.62$; slopes $\approx$ 22.3 vs. 24.7 points/yr), null in 17/18 grid cells — while the same specification rejects at 4/8 placebo cutoffs elsewhere in the series: the instrument demonstrably moves, and reads zero at the release date. The vintage's one new nominally significant result is itself a triage exhibit: a precision-weighted rerun (weights $1/\sigma_i^2$ from

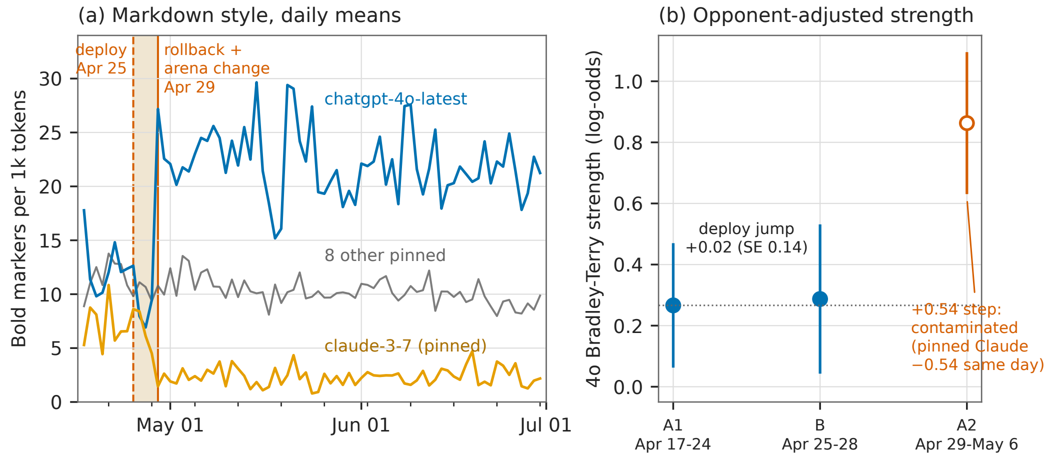

Figure 6: The deploy–rollback episode (reproduced from [47]): (a) daily bold-markers per 1k tokens for 4o, the pinned `claude-3-7` control, and pooled pinned snapshots — the pinned control steps, mirror-imaged, on exactly the rollback date; (b) 4o Bradley–Terry strength by segment: the deploy jump is +0.02 log-odds.

published CIs) returns a negative kink significant in 18/18 cells (-0.0190, $p = 0.016$) — which the battery identifies as smooth post-2024 concavity, not an o1-localized break, because placebo cutoffs at +90/+180 days give same-sign *larger* kinks (-0.0399, $p = 0.0064$; -0.0297, $p = 5.5 \times 10^{-6}$) and a local quadratic absorbs it ($p = 0.14$). A tempting third series — strict frontier records — is a non-design (4 pre-cutoff points in the estimation cell; its nominal $t = 2.28$ dies under a 30-day donut and flips sign under a 45-day one). Whatever the reasoning-model recipe did to AI capabilities, it did not bend the aggregate index's calendar-time trend at the release date (Figure 8).

11 Coverage: everything considered

The survey claim rests on Tables 2 and 3: every dataset–design pair the project considered, with its verdict or its reason for exclusion. Beyond the eight findings above, the studied set includes the flagship graveyard [39] — a $t = 12.9$ "kink" in cumulative GPU-cluster stock at ChatGPT that fails *every* placebo (empirical size 7/7: smooth super-exponential curvature bends everywhere), a sign-unstable Chinchilla null, and the EU threshold correctly rejected as a kink because it is a step with sorting — plus a blind LoRD3 scan that missed Chinchilla's adoption ramp and correctly diagnosed its own top discovery as a fine-tune measurement artifact [43], and a financial-series stress test in which every "credible" automated verdict on semiconductor stocks dissolved under inspection — the DeepSeek "kink" (Holm $p = 1.4 \times 10^{-8}$) is the 2025 AI rally sitting on both sides of a one-week crash [49].

12 Discussion

The scoreboard. Of 17 analyses carried to verdict, four estimates survive as credible (the rewording jump, the legal-channel export bend, the GPQA kink, and — graded moderate — the EU-Act bunching), two nulls carry power certificates (ECI at o1, twice; the LMArena vote null), and at least six nominally significant results were killed or reclassified by the battery: the R1 attribution, the capex kink, the CI-weighted ECI kink, the datacenter $t = 12.9$, the semiconductor

Table 2: Studied to verdict: the 17 dataset–design analyses of record.

Data × design	Cutoff / event	Verdict	Ref.
GPQA-Diamond logit scores × RKD-in-time	o1-preview 2024-09-12	credible, date-localized (placebo 0/7)	[39]
METR 50% time horizon × RKD-in-time	o1-preview	era bend real; date attribution fails (pre-side placebos 3/6)	[39]
ECI 2025 vintage × RKD-in-time	o1-preview	null with power (falsification guard)	[39]
ECI fresh vintage, unweighted / CI-weighted / frontier records × RKD	o1-preview	null upgraded; weighted "kink" = curvature artifact; records series non-estimable	[44]
China vs. hyperscaler chip stocks × group DiK (legal; total incl. smuggled)	controls 2023-10-17	credible (-0.3 to -0.56 ln/yr); total attenuated and fragile	[46]
Country panel, $\geq 10^{24}$-FLOP run counts × SuDDDS DiD + SC/GESS	controls 2022Q4	descriptive-only (placebo-in-space exact $p = 1.0$)	[46]
BTOS sector panel × group DiK + SuDDDS scan	DeepSeek-R1 2025-01-20	descriptive-only; attribution fails triage	[40]
BTOS national spliced series × measurement RDiT + blind LoRD3	rewording, wave 202524	credible; splice factor +6.04 pp / 1.553	[41]
EDGAR capex panel × group DiK	ChatGPT 2022-11-30	descriptive-only; nominal inference identified as artifact	[42]
LMArena battles × ABA ITS + windowed Bradley–Terry	deploy 2025-04-25, rollback 04-29	deploy leg credible (vote null); rollback leg identified arena-side artifact	[47]
Epoch compute panel × binned-Poisson bunching + Fisher contrasts	EU AI Act 10^{25} FLOP, 2024-08-01	credible (moderate): bunching, behavioral or reporting	[45]
EU AI Act threshold × kink/RD probes	10^{25} FLOP	rejected: step with sorting, wrong estimand shape	[39]
GPU-cluster cumulative stock × RKD-in-time	ChatGPT	rejected: placebo size 7/7 on super-exponential series	[39]
Tokens-per-parameter × RKD-in-time	Chinchilla 2022-03-29	sign-unstable null	[39]
Tokens-per-parameter × blind LoRD3 scan	(discovery)	honest miss: top discovery is a fine-tune measurement artifact, correctly diagnosed	[43]
Semiconductor weeklies × kinks, RDD, DiD scan	BIS rounds; DeepSeek week	all "credible" verdicts artifacts of trend regimes; informative DiD null	[49]
ADH Prop-99 panel × blind SuDDDS + synthetic control	1989	benchmark recovered exactly (validation anchor)	[48]

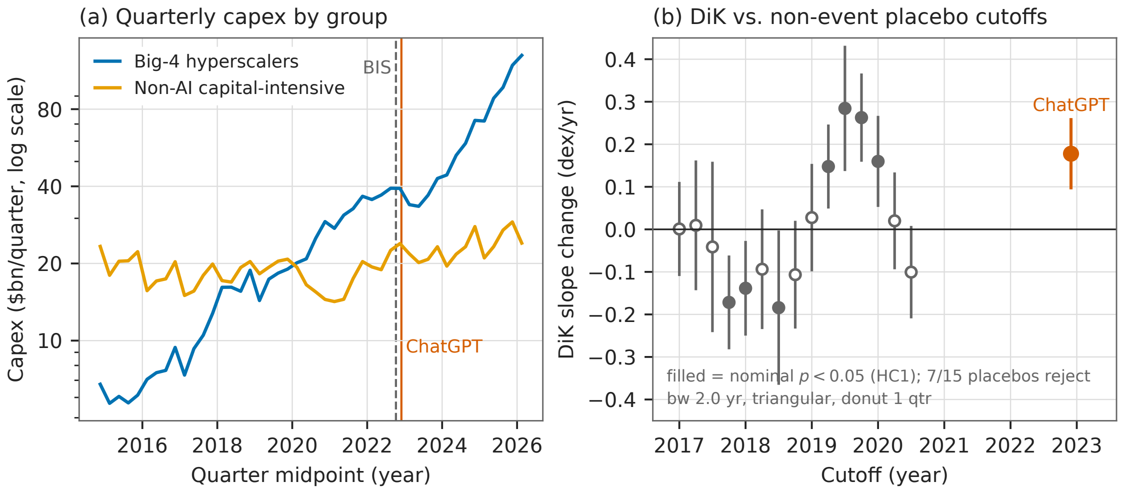

Figure 7: Capex at ChatGPT (reproduced from [42]): (a) quarterly capex of the big-4 and control aggregates with the BIS and ChatGPT dates marked; (b) DiK estimates at 15 pre-period non-event placebo cutoffs versus ChatGPT — seven placebos reject, and the largest placebo $|z|$ exceeds the ChatGPT $|z|$.

"kinks" at three cutoffs, and the LMArena rollback narrative. That kill rate — roughly half of everything that would have cleared a naive $p < 0.05$ bar — is the central empirical fact of the survey.

Nominal inference on the public record of AI is broken by default. The corpus's series are short, serially dependent, and smooth; HC1/CR1 standard errors on them manufacture $|z| > 6$ at dates where nothing happened (capex at 2021.5; datacenters everywhere; the DeepSeek rally kink). Placebo calibration measures the size distortion directly — a nominal 5% test rejecting 47% of the time in the capex pre-period — and is cheap. Every headline in this literature should be read against a placebo distribution or not at all [32, 10].

Era bends are not event bends, and calendar weeks are crowded. The shifted-cutoff grid mechanically separates "the trend changed in this era" (METR, the January-2025 adoption surge) from "this event changed the trend" (GPQA, the rewording). Even when a bend localizes, the week is shared: R1 sits with the diffusion rule, Stargate, and a panel rollover; ChatGPT sits eight weeks from the BIS controls; the export-control kink is localized only to ±1 quarter. Date-localization is necessary, never sufficient, for attribution.

Measurement is a treatment. The single largest causal effect in the corpus — +6 pp of "AI adoption" overnight — was produced by a questionnaire. The EU-Act deficit may partly be labs' disclosure behavior rather than training behavior; the BTOS rollover shifts respondent composition; LMArena's serving configuration moved a pinned model. Instrument changes mimic adoption, compliance mimics avoidance, and platform operations mimic model behavior; designs on the public record need measurement-side placebos (pinned controls, sibling instruments, archived outcomes) as much as treatment-side ones.

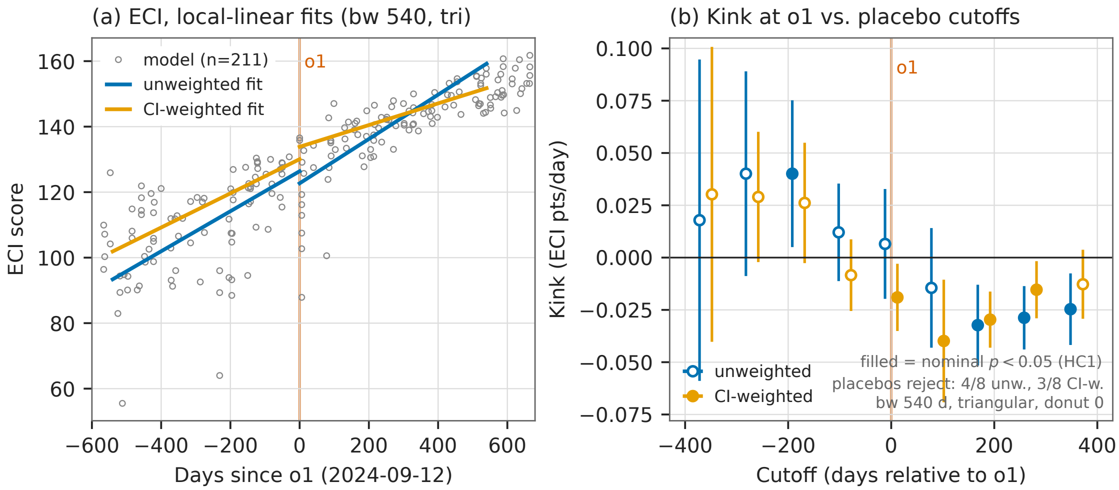

Figure 8: The ECI null (reproduced from [44]): (a) 211 fresh-vintage ECI scores against days since o1-preview with unweighted and CI-weighted per-side fits; (b) kink estimates at the o1 cutoff and eight placebo cutoffs, both weightings — the unweighted design rejects at 4/8 placebos while reading zero at o1.

Nulls and refusals are results. The ECI null is informative because the same specification rejects at 4/8 placebo cutoffs — the dial works and reads zero. The LMArena null is informative because the style break proves the treatment arrived. And the pipeline's refusals (NaN rather than a number when placebo pools are too small) repeatedly prevented fabricated certainty. An automated discovery stack earns trust not by what it finds but by what it declines to claim; the blind Prop-99 recovery, the blind re-localization of the rewording cutoff, and the honestly diagnosed Chinchilla miss are the same machinery validated, confirming, and self-correcting.

Limitations. Calendar-time designs rest on an untestable no-co-located-event assumption that placebo grids probe but cannot certify. Several verdicts lean on few units (4 country groups, 7 sector clusters, 6 quarterly points per cell) where all asymptotic inference is optimistic. Epoch compute figures are partly estimates; benchmark contamination drifts; and every capability result is a property of the release stream, not of individual systems. The survey covers the public record through mid-2026; the queued designs in Table 3 continue it.

Reproducibility

All estimates were produced with the open-source `natex` toolkit [50] (v0.2.0, seeds declared per note) from public data that are never committed to the repository. Every figure in this paper is reused verbatim from a companion note's committed figure, and each of those figures regenerates deterministically from a script that asserts the quoted headline numbers against the analysis records before drawing. The companion notes [39, 41, 46, 40, 42, 47, 45, 44, 43, 48, 49] carry the full specification grids, placebo batteries, and honest-inference notes summarized here.

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Table 3: Considered and excluded: 15 candidate pairs with reasons of record.

Candidate pair	Reason excluded
DCM inference-price series × kink at DeepSeek-V3	series ends 2025-02 with five flat terminal months: no post-slope identifiable
Epoch cyber-vulnerabilities × post-reasoning event study	no processed dataset exists (the published download is the raw CVE git repository)
BTOS state panel × Bartik-style DiD	requires external ACS/BLS exposure weights; ∼40% of state-level cells disclosure-suppressed
R&D input-share table × any family	$n = 7$ rows; no quasi-experimental design applies
GPQA/METR o1 kinks; China DiK × standalone mini-papers	already results of record in [39]; duplication adds nothing
GPU-cluster / datacenter cumulative growth × standalone kink mini-paper	already rejected in the flagship graveyard (placebo empirical size 1.0: curvature, not kink); no standalone treatment
Chip sales + components × chip-level SuDDDS at the Apr-2025 H20 license round	needs a shared chip-alias map and ragged-2026 panel preparation; strongest queued follow-up
Datacenter site-level panel × SuDDDS at Colossus/Stargate entry	18% of timeline rows are unflagged projections requiring filtering; queued
LMArena leaderboard history × RKD around model-spec releases	cumulative Bradley–Terry ratings are serially cumulative series; the GPU-cluster placebo failure is the direct precedent
LMArena style-premium DiD around Model Spec / constitution updates	spec-with-model-update confound requires judge-score validation not yet run
AI-company revenue/WAU × kink at ChatGPT	zero pre-cutoff mass at the treatment date; not formalizable
Polling cross-section × threshold RDs (income step, age cliff)	single survey wave; no microdata, sample sizes, or field dates
Values-in-the-wild / CCAI votes × any design	one aggregate snapshot; no behavioral treatment channel
Benchmark panel × teaching-to-the-test DEE with IRT difficulties	requires a 52-benchmark IRT panel build; queued
All-models panel × bunching at 10^{26} FLOP (US EO line) + disclosure-as-outcome scan	only 3 models ever above the line; no identifiable density contrast

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